

Algorithm to Automatically Detect Abnormally Long Periods of Inactivity in a Home

Paul Cuddihy, Jenny Weisenberg, Catherine Graichen, Meena Ganesh
General Electric Global Research
One Research Circle
Niskayuna, NY, USA
+1 518 387 5000

{cuddihy, weisenje, graichen, ganeshme}@research.ge.com

ABSTRACT

An algorithm has been developed to automatically construct individual models of normal activity within a home using motion sensor data. Alerts can be generated when a period of inactivity exceeds a normal length for a particular residence. Alerting frequency has been optimized on a total of 650 days of real data from four homes of seniors who live independently. Results suggest that an inexpensive system that does not require the occupant to push any buttons or wear any devices could nonetheless alert within hours if a senior is unusually inactive. Further, such algorithms may facilitate widespread deployment of smart home technology to persons with different behavior patterns and home layouts by using automatic learning in place of potentially tedious manual configuration.

Categories and Subject Descriptors

J.3 [LIFE AND MEDICAL SCIENCES]: Health

General Terms

Algorithms, Experimentation, Human Factors

Keywords

Activity detection, outlier detection, automated learning, elder monitoring, motion sensing.

1. INTRODUCTION

In countries around the world, the proportion of elderly in the population is growing at a rapid rate. Many of these seniors prefer to live independently for as long as they are able, despite the onset of conditions such as frailty and dementia. Solutions are needed to enable independent living while enhancing seniors' safety and their families' peace of mind.

One of the foremost worries on the part of seniors and their

families is that of a senior being immobilized due to, for example, a fall, and not being able to get up or summon assistance [1]. In this situation, people can be left waiting in an immobilized state for long periods of time before a neighbor or caregiver becomes aware of the situation [2]. Solutions to this problem currently include worn devices such as personal emergency response systems (PERS), where the user can push a button on a pendant to initiate a speakerphone conversation and summon help.

Many researchers in business and academia have set up smart homes to study behavior patterns in support of independent living [3] [4]. Remote monitoring solutions for the elderly have been reported widely and are in the early stages of adoption [5]. Research has also been done in the area of fall detection using impact and orientation detection [6].

However, solutions typically require intervention by the resident and caregiver, in the form of the resident wearing or carrying a device, pressing a button for help, or the caregiver manually configuring a system with time limits.

Here, we focus on a system to detect unusually long periods of inactivity, while requiring neither intervention by the resident nor the manual configuration of limits by the caregiver [7]. Instead, a small number of wireless motion sensors and door sensors were deployed in the home, collecting data to model the resident's normal activity. These sensors were considered a good choice because they are familiar fixtures in home security systems and are relatively inexpensive. Raw data is collected from the sensors and is used to draw conclusions about the resident's activities on a daily basis. This information is then made available to their caregivers via a secure website. An important part of the system is the ability to receive alerts when a potentially troublesome situation occurs. For example, an unusually long inactive period in the home could signal a possible fall or other trouble.

The Automatic Inactivity Detection (AID) system has been designed to automatically adjust to the configuration of the sensors and the activity patterns of the resident [8]. After 10 days of learning, it will begin alerting when unusually long periods of inactivity occur in the home. The algorithm parameters have been optimized to produce less than one alert per month—with a slightly higher rate during the early weeks of learning. This frequency of alerting was determined to be appropriate for family caregivers who are interested enough in unusual inactivity to review alerts occasionally without considering them “false alarms”.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee.

HealthNet'07, June 11, 2007, San Juan, Puerto Rico, USA.

Copyright 2007 ACM 978-1-59593-767-4/07/0006...\$5.00.

Table 1. Inactivity data collected in 30-min intervals

Date	0:00	...	6:00	6:30	6:00	7:00	7:30	8:00	8:30	9:00	9:30	10:00	10:30	...	23:30
12/1	96	...	251	281	311	25	0	14	44	0	15	3	33	...	42
12/2	52	...	412	442	472	502	12	0	0	2	16	8	0	...	-1
12/3	-1	...	-1	-1	-1	-1	-1	-1	-1	-1	0	2	32	...	45

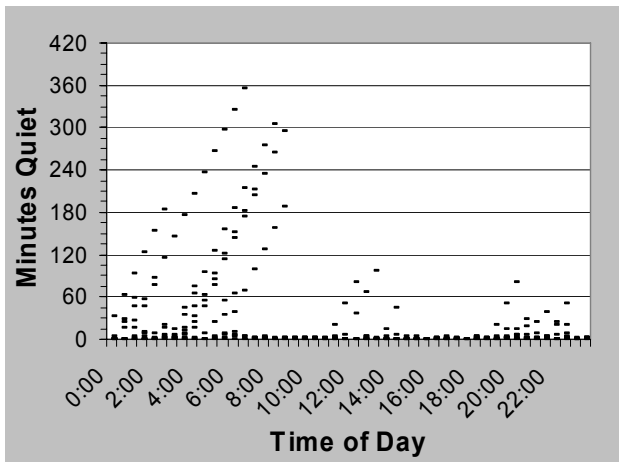
After optimizing the alert frequency, we explore the sensitivity that can be achieved with this relatively low cost, unobtrusive system. The results are promising: roughly equivalent to a vigilant neighbor or family member checking in several times per day. Finally, we discuss ideas on how sensitivity might be increased to approach the requirements for a fall detection system.

2. THE MONITORING SYSTEM

Homes were outfitted with 6 to 9 wireless GE passive infrared (PIR) motion sensors commonly found in home security systems. Sensors were placed near outside doors and in the most commonly occupied living areas around the home. Some participants chose not to have a motion sensor in the bedroom. In all cases, the route from the bedroom to the bathroom was covered. No residents in this study owned pets. However, the motion sensors have pet immunity features and could have been configured to ignore pets up to 40 lbs.

External doors were monitored by wireless magnetic reed switches. As expected of a real-world application, the sensor placement varied by home layout and preferences of the resident.

Data was transmitted out of the home for off-site processing and alert generation. In a final application, alerts could easily be delivered to caregivers via phone, email or the Internet.

**Figure 1. Inactivity data for several days**

2.1 The Alerting Algorithm

The AID algorithm is based on inactivity duration data from the home over the past weeks and months. This historical inactivity data, as seen in Table 1, consists of the elapsed duration of time that no activity has been seen in the home. This data is recorded for each 30-minute interval of time during the day. Each data point represents, for the given time of day, the number of minutes since activity was last detected in the home.

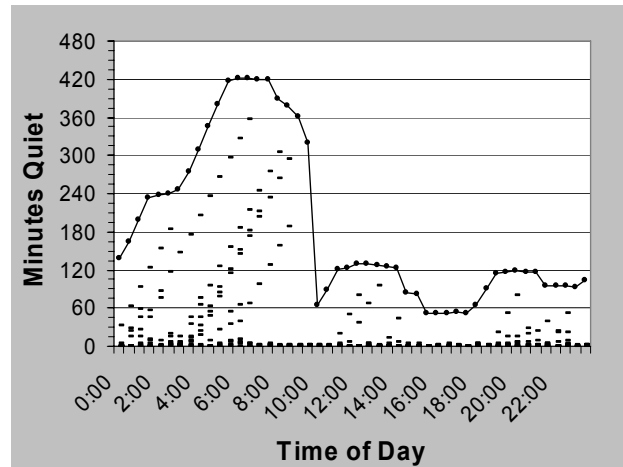
Figure 1 shows several days of inactivity data in a scatter plot. Each point on the graph corresponds to the elapsed inactive time recorded at the given time during one of the days.

From this graph we can see that this household is typically inactive between midnight and 7:00am. Additionally, there are periods of inactivity around the noon hour and in the late evening. No inactivity events are recorded in the late morning or afternoon.

The ability to detect when the senior has left the home is crucial to being able to use AID to detect abnormal inactivity events. Accepting times away from home as normal inactivity would degrade the sensitivity of the system. In the worst case, consider a day when the senior spent a night away from home, returning at 1:00pm. If this were accepted as normal inactivity, the system would no longer be able to detect a late wake-up until after 1:00pm. Instead, the away time must be treated similar to active time. Elapsed inactive times indicating that the senior is away from home are coded as -1.

2.2 Generating an Alert Line

The AID algorithm works by using the elapsed inactivity data to create a threshold for acceptable elapsed inactive time for each time of day, in 30-minute increments. If the resident is inactive for longer than the determined duration, then an alert should be sent to the caregiver. The 48 alert points, one for each time interval, form an alert line that can be superimposed on the elapsed inactive data, as shown in Figure 2.

**Figure 2. Inactivity data with alert line**

One strategy might be to alert at a certain percentile of the historical data for each time interval. A 100th percentile line, for example, would overlap with the longest historical elapsed

Table 2. Configurable parameters to the AID algorithm

Parameter	Function	Range
Maximum Percentile (MP)	Used to remove from consideration the outlier elapsed inactive times from historical data. The lower the value for MP , the more historical data is eliminated from being considered as part of the baseline data.	0.9 - 1.0
Uniform Buffer Percentage (UBP)	Used to determine the size of the uniform buffer to be applied to construct the alert line.	0.2 - 0.4
Variable Buffer Percentage (VBP)	Used to control how much typical activity level affects the alert line. If VBP is high, then the algorithm becomes more sensitive at typically active times, and less sensitive at typically inactive times.	0.2 - 0.5
Interval Weights (W_r for $-2 \leq r \leq 2$)	Controls how much to consider data from the current intervals and the two intervals preceding and following it. Index r represents the interval relative to the current interval. For example, if W_{-1} is the interval preceding the current interval.	1 - 5

inactive time for each interval. It would alert any time a new inactivity period occurs which is longer than any seen in the past.

This over-simplified approach reveals several challenges in analyzing the activity patterns of humans. First, a resident sleeping 5 minutes longer than usual is unlikely to constitute an emergency. It is thus necessary to allow for a buffer zone between the historical data and the alert line, ensuring that an alert is not issued until the resident is substantially more inactive than usual.

Similarly, it is important to provide for modest shifts to a resident's schedule. For example, a person who typically sleeps from 11:00 to 8:00 should probably be allowed to instead sleep from 11:30 to 8:30 without causing an alert. This necessitates a form of lateral buffering to ensure that slight shifts in the person's activity do not trigger excessive alerts.

Further, the size of the buffer applied above the historical data needs to be proportional to the data itself. A time interval with historical data never exceeding 30 minutes can tolerate a smaller buffer than a time interval during which the resident is routinely inactive for up to 8 hours. Also, in most cases we would like to avoid alerting when a resident sleeps more soundly than usual. However, applying a buffer calculated as a simple percentage of the data would mean that active times of the day (where data is near 0) would receive a negligible buffer, resulting in a high rate of false alerts in many homes. Thus, a minimum buffer is needed to ensure that even very active times are granted some leeway.

2.3 Final Algorithm

The AID algorithm addresses these challenges by generating an alert line using the parameters described in Table 2. The alert line is generated by taking all but the very longest historical elapsed inactive times and adding a buffer which is proportional to the size of the inactive data but also guarantees a minimum buffer for every interval.

The formulae used to compute the alert point for each time interval are described below. Index i refers to the time interval for which an alert point is being computed, for example index $i = 44$ represents 21:30. There are three components that make up the alert point for a given interval as shown in equation (1): a high

normal elapsed inactive time, a uniform buffer, and a variable buffer. These components are further described below.

$$ALERT_i = M_i + UB + VB_i \quad (1)$$

Each of the components requires the computation of a high percentile of the historical elapsed inactive data for the time interval. Configurable parameter MP is the percentile to be used for this purpose. Using a percentile of data serves to eliminate from the historical data the very longest historical periods of inactivity for each time interval, such that the definition of typical data is not unnaturally inflated by outlier inactivity data. In equation (2), $data_i$ represents the historical elapsed inactive times for the interval i .

$$m_i = PERCENTILE(data_i, MP) \quad (2)$$

The m_i values will be used when computing each of the three components below.

The first component, M_i , serves to approximate a maximum normal elapsed inactive time for this interval. It considers preceding and following intervals as shown in equation (3), so as to smooth the alert line, providing a lateral buffer. This component is computed as the maximum of the earlier computed percentile values, considering only the current interval and the two intervals preceding and following it. Note that preceding and following interval indices are computed modulo 48 (the number of time intervals) since they represent a daily pattern.

$$M_i = MAX(m_{i-2}, m_{i-1}, m_i, m_{i+1}, m_{i+2}) \quad (3)$$

The second component, UB , represents a uniform buffer to be applied at every interval. It is calculated using equation (4). Configurable parameter UBP is used to control the size of this uniform buffer.

$$UB = UBP \times MAX(m_i) \quad (4)$$

The third component, VB , serves to increase sensitivity during typically active times, and decrease sensitivity during typically inactive times. It uses a weighted average across the current,

preceding and following intervals. Configurable parameters VBP and W_r are used to compute this component as shown in equation (5).

$$VB_i = \frac{1}{\sum W_r} \left[\sum_{r=-2}^2 (VBP \times m_{i+r} \times W_r) \right] \quad (5)$$

After the alert point $ALERT_i$ is computed for each interval, it can be used to trigger alerts in future data. If an elapsed inactivity duration for interval i exceeds $ALERT_i$, then an alert should be issued. The optimal value for configurable parameters MP , UBP , VBP , and W_r will be discussed below.

The algorithm is designed to operate on a rolling window of data, enabling it to adapt to changing resident behavior. Every time the alert line is re-calculated, the oldest data is removed from consideration, while the most recent data is added. A suitable rolling window must be short enough to adapt to changes in resident behavior, but long enough to provide stability to the alert line and to avoid an undue level of false alerts. Using data from our homes, we performed alert simulations on rolling windows of size 15, 30, 45, and 60 days. We concluded that rolling windows of size 60 and 45 days performed similarly in terms of alert line and alert rate, while rolling windows of fewer than 45 days degraded performance by causing a high number of false alerts. The 45 day rolling window was therefore chosen as the shortest rolling window that provided acceptable performance.

2.4 Removing Training Outliers

The AID method works by establishing baseline behavior for the home, typically requiring 10 days of training, during which data is collected but no alerts are generated. (A complete monitoring system may provide other non-adaptive, non-learning alerting capabilities on inactivity during this training period.) It is possible that during this training time, the resident may exhibit abnormal behavior, or a sensor may malfunction. The system must be made robust to these kinds of abnormal occurrences during the training period.

For this reason, the method incorporates a technique to remove outlying data points from the 10 training days. It works by identifying outliers to the overall dataset, as well as data points that are outliers within individual intervals of time. When an outlier data point is identified, then all the data values leading up to it that represent the same inactive period are also removed from the data set.

Note that we refer to these initial 10 days as the “training period” because this is when the initial baseline is created. However, due to the use of the rolling window, the algorithm is continually adjusting and re-training as new data arrives.

3. OPTIMIZING ALERT ACCURACY

The AID algorithm was tested and optimized on a set of data collected from the homes of 4 seniors. In total, 658 days of motion sensor data were used. Door contact sensors made it possible to detect periods during which the resident was away from home.

All proposed algorithms were tested on the raw sensor data exactly as it occurred. However, the data was also used more fully by considering how the algorithm would perform if the same data were collected with days in a different order. For example,

the first 10 days of raw data from each home would, in reality, be used as training data. However, given the rest of the data, we were able to simulate what would have happened had any consecutive 10-day subset comprised the initial training period. In order to make maximal use of the motion sensor data collected, 256 permuted datasets were constructed using real subsets of consecutive days of the raw data. To minimize discontinuities across the midnight boundaries, the subsets for the 10-day training set, the post-training learning period and the stable test period use consecutive days within the subset. However, discontinuities between these subsets occur through the permutation process. These two discontinuities for each test case—most likely occurring during sleep times—should have minimal impact.

3.1 Performance Criteria

The goal of the AID algorithm is to trigger an alert when the resident is inactive in the home for an abnormally long period of time. A reasonable balance must be achieved between producing false alerts due to high sensitivity, and missing a valid alert due to low sensitivity. The performance criteria were defined with this tradeoff in mind. There are two types of performance criteria – the first aims to ensure that the total alert rate (and thus the false alert rate) is kept reasonably low, while the second type aims to ensure that the algorithm is sensitive enough to in fact trigger an alert in an acceptable amount of time after a long inactivity event occurs.

Controlling the alert rate is of primary importance, since the user will reject a system that creates a large number of false alerts. As the system learns the resident’s typical behavior during the first 10 days after installation, no alerts will be triggered. After this training period ends, the system will be usable but may be prone to a higher than usual number of alerts as it continues to learn the resident’s behavior. After the post-training period is over, the system should be stable and alert infrequently. The performance criteria chosen for this alert rate are shown in Table 3.

Table 3. Performance criteria

Criterion	Objective
Alert rate in 21-day post-training period	Maximum average 1 alert per week
Alert rate during normal operation	Maximum average 1 alert per month
Time to Alert	Minimize

The second objective is to ensure that the system is sensitive enough to trigger alerts within a reasonable amount of time after an abnormal inactivity event occurs. To gauge this sensitivity, we measure the positioning of the alert line. A low alert line means a relatively short time to alert, and an algorithm that is more likely to alert should a period of inactivity occur. Reports from caregivers during the study indicated that our data contained no unusually long inactive periods caused by illness or falls. Therefore, we rely on the positioning of the alert line to indicate how long the resident *would have* needed to be inactive in the home before triggering an alert.

3.2 Results

A Java program was developed to generate the permuted datasets from the raw data, run the AID algorithm on each permuted dataset, and report results. We optimized the algorithm by running this program using various values of the configurable parameters. A design of experiments approach was used to determine the best combination of settings for each parameter, within the ranges specified in Table 2. Reasonable settings for these parameters were found to be $MP=0.97$, $UBP=0.30$, $VBP=0.40$, and $W_r = \{1,2,3,2,1\}$.

Using the optimized parameters, 93% of the 256 runs met the performance objective of one or fewer alerts per week during the 3-week post-training period. 91% met the objective of one or fewer alerts per month thereafter. On average, a dataset triggered less than 1 alert (0.79) in the entire 3-week post-training period, and less than 1 alert (0.40) in the 6-week normal operation period.

The sensitivity is plotted in Figure 3. The chart is constructed by reordering the time intervals of each alert line in ascending order by the alert threshold, so point 1 represents the percentile of smallest alert threshold over all time intervals over all test sets. The “hockey stick” curve corresponds closely to the area of the alert line associated with typical sleep hours (eight hours of nighttime) during which the algorithm is less sensitive. During the day, most residents will generate an alert if inactive for several hours. At night, the system typically does not alert unless the resident misses their usual wake-up time by a substantial amount of time.

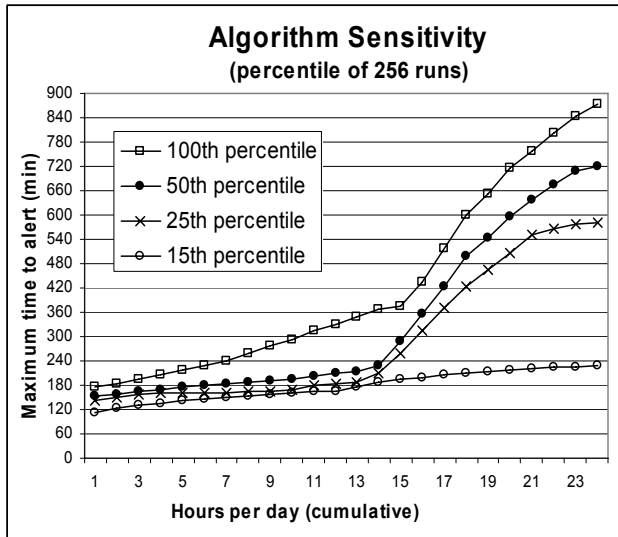


Figure 3. Sensitivity of optimized algorithm

4. DISCUSSION

The system appears to reasonably simulate the kind of vigilance achieved by a neighbor or family member checking in several times a day. An unusual level of inactivity is detected within several hours.

With decreased sensitivity during long nighttime periods of inactivity, the system is unlikely to wake a caregiver to report that a resident is sleeping unusually well. The price paid for this feature is that a fall at bedtime may not be detected until morning. We anticipate that occupancy sensors in the bed and frequently

used chairs could mitigate this type of low sensitivity and possibly raise the overall sensitivity of the system close to what might be expected in a fall detection system.

The vast majority of simulation runs met the alerting performance criteria. Inspection of those that alerted too frequently suggests that the system could benefit from simple caregiver feedback on some alerts, such as identifying certain inactive times as normal or allowing the user to dial overall sensitivity up or down.

The system exhibits some attractive features, including relatively low cost; unobtrusiveness to the resident; little or no configuration of alerts, and the ability to adapt to different homes, residents, and sensor layouts in only 10 days of training.

It should be noted that the AID algorithm as used here is monitoring the activity level of a home and not of a single person. One of the test homes was occupied by a married couple. Anecdotally, it appears that the increased activity generated by two people resulted in faster alerting times and more accurate determination of when the resident is away from home. The system works on the presumption that as long as one person is moving, they are capable of reporting any trouble in the home. Clearly this assumption can not be applied to a couple where one member suffers from dementia, or any cognitive or physical condition that renders them unable to summon help on their own.

We received informal feedback from the users of this system. Most of the participating caregivers felt that the system increased their peace of mind in caring for their parent. Some users logged into the website on a daily basis to check on their parent's activities, while others relied on the alerts, such as the one described in this paper, to be generated when appropriate. At least one elderly participant reported feeling more comfortable with this system in place, although a second elderly resident voiced a discomfort with having his movements monitored in his home. Users of the system submitted a number of requests for future enhancements, such as incorporating medication alerts, temperature sensors, and car sensors.

5. CONCLUSIONS

Using motion sensor data, the AID algorithm can detect unusual periods of inactivity in a senior's home with a level of sensitivity similar to a family caregiver checking in several times a day. The use of motion sensors for this purpose is inexpensive and requires no wearing or action on the part of the senior. Alerting requires little or no configuration for individual homes, sensor layouts, or behavior patterns. Instead, this method trains in 10 days and has a reasonably low rate of false alerts.

The sensitivity could likely be improved by the addition of occupancy sensors in areas where the resident spends long periods of inactivity, such as the bed and frequently used chairs.

6. FUTURE WORK

The addition of bed and chair occupancy sensors to improve sensitivity during the night should be investigated. A bed occupancy sensor could generate a signal that the AID algorithm that the resident is in a known location. This would greatly improve sensitivity for a fall event near bedtime. Bed and chair occupancy sensors would need to have pet immunity features such as requiring a minimum weight change of 50 lbs before the sensor fires.

A system implementation of this type would also require a second instantiation of the alerting algorithm to warn when a resident has stayed in bed (or another location with an occupancy sensor) too long. This approach may be used in a favorite chair, and other areas of the home that are safely occupied at low activity levels for long periods of time.

The practical application of this algorithm requires robust determination of when the resident is away from home, as mis-identifying these times as normal inactivity will seriously degrade sensitivity. For this research, the use of magnetic reed switches on the doors in combination with the motion sensors provided fairly accurate determination of when the resident was away from home. Inaccurate results were occasionally caused by problems such as sensor coverage, lack of precision in the times reported for the beginning and end of motion, and visitors to the home. These problems all seem surmountable, but such a system remains to be built and demonstrated on a large data set.

Some behavior patterns may exhibit different cycles, such as a weekly pattern. The implementation of the AID algorithm may be configured to use time intervals over a weekly period rather than a daily period. However, this requires a longer elapsed period to collect sufficient historical data for weekly analysis.

Other areas of interest include allowing the caregiver to provide simple feedback on alerts and overall sensitivity in order to improve the speed and effectiveness of training. Formal investigation into the optimal number and placement of sensors will help maximize the cost-effectiveness of the system.

It is possible to apply the AID algorithm in a variety of scenarios other than looking for inactive periods using motion sensors. Caregivers may be interested in receiving alerts on additional events such as failure to use the bathroom for too long, or being active for too long. The AID algorithm is designed to be adaptive to a wide variety of elapsed duration data. Clearly, it will need to be optimized differently for these different uses.

Alternatively, techniques such as hidden markov models, expectation maximization clustering, and parametric statistical techniques could be re-examined for more complex alerting tasks. Our experiments with these techniques have suggested that they struggle to meet the requirement of learning enough from the first ten days of data to begin alerting. In order to build large enough

training sets, it may be necessary to devise an algorithm that uses the first ten days of data to identify additional similar homes. Data from the target home along with other similar homes might then be sufficient to train these types of models and achieve reliable alerting after only ten days.

7. ACKNOWLEDGMENTS

We would like to thank Lisa Gaudet, Director of Remote Care Technology and Genetic Services at Northeast Health for recruiting some of our study participants. We also thank the study participants for allowing data to be collected from their homes.

8. REFERENCES

- [1] Tinetti, M.E., Liu, W.L., and Claus, E.B. Predictors and prognosis of inability to get up after falls among elderly persons. *JAMA*, Vol 269, No 1 (Jan. 1993), 65-70.
- [2] Wild, D., Nayak, U.S.L., Isaacs, B. How dangerous are falls in old people at home? *British Medical Journal*, Vol 282 (Jan. 1981), 266-268.
- [3] Guralnik, V., Haigh, K.Z. Learning Models of Human Behaviour with Sequential Patterns. AAAI-02 Workshop "Automation as Caregiver", pp 24-30, 2002.
- [4] Mynatt E.D., Essa, I., and Rogers, W. Increasing the Opportunities for Aging-in-Place. *Proceedings of ACM Conference on Universal Usability*, pp 65-71, 2000.
- [5] Wallace, D.J. Gadget Knows Grandma Inside Out. *The New York Times*, Oct 23, 2003.
- [6] Williams, G., Doughty, K., Cameron, K., and Bradley, D.A.. A Smart Fall & Activity Monitor For Telecare Applications. In *Proceedings of the 20th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, Vol 20, No 3, 1998.
- [7] Cuddihy, P.E. Successful Aging: Home Assurance System. *IEEE Pervasive Computing*, Vol 3, No 2, 2004.
- [8] Cuddihy, P.E., Ganesh, M., Graichen, C.M., and Weisenberg, J.M. US Patent #7091865. System and method for determining periods of interest in home of persons living independently. Patent granted August 2006.